Batch-Invariant LLM Serving on Hopper Without Custom Kernels

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Abstract

[USER: write the abstract. Points it should cover: (1) served LLM outputs depend on batch neighbors and prefix-cache state; (2) vLLM's batch-invariant mode fixes this at a large throughput cost on models without tuned kernels; (3) on Hopper and Blackwell, cuBLAS is already batch-invariant under the flags that mode sets, but Linear layers are still routed to a Triton kernel; (4) routing them to cuBLAS keeps outputs identical at a small fraction of the cost; (5) limits: Gemma-2, the TP=2 overhead, tuned models.]

1 Introduction

[DRAFT for the user to rewrite; framing is the user's.] Greedy decoding is often assumed to be deterministic, but in a batched inference server it is not. The same request, served alone or alongside unrelated requests, with or without a cached prefix, can produce different tokens, because the floating-point reduction order of matrix multiplication and attention kernels depends on the shape of the batch [He and Thinking Machines Lab, 2025, Yuan et al., 2025]. In our measurements on vLLM [Kwon et al., 2023], 41 of 100 greedy math requests to Qwen2.5-7B-Instruct changed tokens across these conditions.

vLLM offers a batch-invariant mode that removes this dependence by replacing reduction kernels with fixed-order implementations. On an H100 it reduced decode throughput at batch 32 by 62.3% for Qwen2.5-7B-Instruct. vLLM has since added hand-tuned kernel configurations for a few models [vLLM contributors, 2026], which recover most of the loss for those models only.

We observe that on Hopper (SM90) and Blackwell (SM100) GPUs, vLLM's batch-invariant mode already configures cuBLAS on the premise that split-K is its only source of batch variance, yet still routes every unquantized Linear layer through its Triton kernel. Routing these layers to cuBLAS instead keeps outputs bitwise identical across serving conditions and removes nearly all of the cost. We contribute:

- a direct bitwise test showing that cuBLAS matrix multiplication is batch-invariant under batch-invariant mode's flags on H100 and B200, across 4 cuBLAS versions (1,154,880 row comparisons, 0 mismatches);
- end-to-end serving tests over tensor parallelism, long context with chunked prefill, seeded sampling, and four model families, with preregistered pass/fail rules and controls;
- throughput measurements against vLLM's own batch-invariant mode on current vLLM main;
- a diagnosis of the two places it falls short: the TP=2 overhead comes from fixed-order communication, and a Gemma-2 failure lies in vLLM's FlashAttention path, not in the routing.

2 Background

A batch-invariant kernel computes each row of its output with the same sequence of floating-point operations regardless of how many other rows are in the batch. Reductions whose split depends on batch size (split-K matrix multiplication, split-KV attention, and some normalization kernels) break this property [He and Thinking Machines Lab, 2025]. vLLM's batch-invariant mode, enabled with `VLLM_BATCH_INVARIANT=1`, pins attention to a single split, replaces reductions with fixed-order kernels, disables reduced-precision reductions, prefers cuBLASLt, and pins collective communication to fixed algorithms. On Ampere (SM80) it overrides matrix multiplication with a persistent Triton kernel, since cuBLASLt there cannot be made batch-invariant. On SM90 and SM100 it instead sets cuBLAS workspace variables, with a source comment stating that split-K is the only source of batch variance on those GPUs. Yet `UnquantizedLinearMethod.apply` calls the Triton `linear_batch_invariant` on every CUDA device, so on Hopper the Linear layers, which perform most of the matrix-multiplication work, still run the Triton kernel.

3 Method

The change is one routing decision: under batch-invariant mode on SM90 and newer, Linear layers call `torch.nn.functional.linear` (cuBLAS) instead of the Triton kernel. Everything else in batch-invariant mode is unchanged. Our prototype applies this as an import hook that patches every vLLM process, including tensor-parallel workers, and logs a marker per process; a run counts only if the expected markers are present. An upstream version should also run a short bitwise self-test at startup and fall back to the Triton path if cuBLAS ever fails it, since NVIDIA does not document this behavior as a guarantee.

4 Evaluation

Setup. vLLM main (0.33.0rc1.dev454), PyTorch 2.14 with CUDA 13.0, bf16, NVIDIA H100 80GB unless noted. Determinism: each request is served 8 times in random order under four conditions (cold or warm prefix cache, alone or with 7 unrelated requests, each twice), and every pair of the 8 outputs is compared token by token; top-20 logprobs are also recorded. Prompts are few-shot GSM8K and MATH-500 problems. Pass/fail rules were written before each run (repository `PROTOCOL.md`); a control with batch-invariant mode off must diverge for a test to count.

Is cuBLAS batch-invariant? We multiply each row alone and inside batches of 1 to 8192 rows, at every Linear shape of six models, in bf16 and fp16, and compare rows near the start and end of the batch (Table 1). Under batch-invariant mode's flags, no row differed on any of 7 H100 builds or 2 B200 builds, including with a second process saturating the GPU. Setting only the PyTorch flags, without the cuBLAS environment variables, made no difference. With PyTorch defaults, between 6,784 and 8,544 of 38,496 rows differed on H100, and 19,544 on B200, so the test detects batch dependence when it exists.

End-to-end determinism. With Linear layers on cuBLAS, 6 of 7 serving configurations produced identical tokens and identical top-k logprobs across all conditions (Table 2). Controls with batch-invariant mode off diverged on 4% to 76% of requests. Gemma-2-9B fails: 39 of 100 requests diverged with cuBLAS routing, and 40 with vLLM's stock batch-invariant mode, against 60 with the mode off. vLLM's batch-invariant mode does not cover Gemma-2 in this setup; the routing

GPU, torch build	BI flags	Flags only	Defaults
B200, torch t2.13.0+cu130	0	0	19,544
B200, torch t2.14.0+cu130	0	0	19,544
H100, torch t2.11.0+cu128	0	0	7,344
H100, torch t2.12.0+cu126	0	0	6,784
H100, torch t2.13.0+cu126	0	0	6,784
H100, torch t2.13.0+cu129	0	0	7,344
H100, torch t2.13.0+cu130	0	0	8,544
H100, torch t2.14.0+cu126	0	0	6,784
H100, torch t2.14.0+cu130	0	0	8,544
H100, torch t2.14.0+cu130 + GPU load	0	0	8,544

Table 1: Rows that differ between a row multiplied alone and inside a batch, out of 38,496 per cell. BI flags: environment variables and PyTorch flags as batch-invariant mode sets them. Flags only: PyTorch flags without the cuBLAS environment variables. Defaults: PyTorch defaults (control).

Configuration	Plain cuBLAS	Mode off	Markers
Qwen2.5-7B, TP=2	0 / 100	38 / 100	2
Qwen2.5-7B, TP=4 (TP=2 control)	0 / 100	38 / 100	4
Qwen2.5-7B, long context, chunked prefill	0 / 50	29 / 50	1
Qwen2.5-7B, seeded sampling	0 / 100	40 / 100	1
Qwen2.5-32B, TP=2	0 / 100	76 / 100	2
Mistral-7B	0 / 100	4 / 100	1
Gemma-2-9B	39 / 100	60 / 100	1

Table 2: Requests whose 8 runs were not all identical, with Linear layers on cuBLAS under batch-invariant mode (plain cuBLAS) and with batch-invariant mode off (control). Markers count processes that loaded the routing change.

change neither causes nor fixes this. Varying one component at a time under stock batch-invariant mode (30 requests each) localizes the failure: switching attention to the Triton backend removes it (0 diverging requests, against 19 for its control), as does eager execution (0, against 23). Removing logit softcapping does not (15, against 24), and Gemma-7B, which shares Gemma-2's head size but has no softcapping, is unaffected (0, against 30). The failure therefore lies in FlashAttention combined with compilation or CUDA graphs on Gemma-2. [Follow-ups pending: sliding window disabled; CUDA graphs off with compilation kept.]

Throughput. Table 3 reports decode throughput loss at batch 32 against batch-invariant mode off, the median of three independent jobs. For models without tuned configurations, stock batch-invariant mode loses 63.1% (Qwen2.5-7B) and 59.2% (Gemma-2-9B); cuBLAS routing loses 0.2% and 1.0%. Mistral-7B shares most Linear shapes with Llama-3.1-8B, which vLLM has tuned, so stock mode is already cheap there (5.8%). For prefill, cuBLAS routing loses at most 5.0% against 9.5% to 25.5% for stock mode. 16 of 17 measured cells meet our preregistered limit of 5%. The exception is Qwen2.5-7B at TP=2, which loses 5.8% (stock: 61.0%). This excess comes from batch-invariant mode's fixed-order communication settings, not from the routing change: applying only those settings, with batch-invariant mode otherwise off, costs 6.0% on its own. Cheaper deterministic communication is a separate target.

[Optional section: effect on machine translation outputs and metrics.]

Model	Stock	Plain cuBLAS	Plain, worst repeat
Qwen2.5-7B	63.1%	0.2%	3.1%
Gemma-2-9B	59.2%	1.0%	1.3%
Mistral-7B	5.8%	3.3%	5.9%
Qwen2.5-7B, TP=2	61.0%	5.8%	6.8%
Qwen2.5-32B, TP=2	58.2%	−2.3%	−1.0%

Table 3: Decode throughput loss at batch 32 against batch-invariant mode off. Median of three jobs.

5 Limitations

The bitwise property is observed, not guaranteed by NVIDIA, which is why an upstream change should include a startup self-test. We tested one engine, single-node tensor parallelism up to four GPUs, dense models up to 32B parameters, and bf16; we did not test H200, multi-node serving, mixture-of-experts models, quantized models, or production traffic. Gemma-2-9B is not batch-invariant under vLLM's batch-invariant mode with its default FlashAttention backend. For models in vLLM's tuned table the benefit is small. We compare only against vLLM's own batch-invariant mode, not against other engines' deterministic modes or LLM-42 [Gond et al., 2026].

6 Related Work

He and Thinking Machines Lab [2025] identify batch-size-dependent reductions as the main source of inference nondeterminism and describe batch-invariant kernels; vLLM's batch-invariant mode follows this design. Yuan et al. [2025] show that numerical nondeterminism changes reasoning accuracy across batch sizes, GPUs, and precisions. LLM-42 [Gond et al., 2026] obtains determinism through verified speculation rather than invariant kernels. Our contribution is narrower: on current NVIDIA hardware the vendor matrix-multiplication library already satisfies batch invariance under the right flags, so the most expensive custom kernel is unnecessary there.

7 Conclusion

[USER: conclusion framing.] Code, raw outputs, and the decision rules for every experiment are at `https://github.com/imthomp/vllm-batch-invariance`.

References

Raja Gond, Aditya K. Kamath, Ramachandran Ramjee, and Ashish Panwar. LLM-42: Enabling determinism in LLM inference with verified speculation, 2026.

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vLLM contributors. [kernel][perf] add H100 tuned configs for Llama-3.1-8B and Qwen3-8B to the batch-invariant matmul table. vLLM pull request 62219, 2026. URL https://github.com/vllm-project/vllm/pull/62219.

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